PH LABS

Research Compound Protocol Library

28-page laboratory reference for in-vitro research

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Table of Contents

A 28-page laboratory reference covering handling, reconstitution, storage, quality assurance and compound-specific research protocols.

Chapter 1

About This Library

Purpose, audience and how to use this document

The PH Labs Protocol Library is a laboratory reference for researchers working with our HPLC-verified research compounds. It consolidates handling, reconstitution, storage, and quality-assurance procedures into a single document so that principal investigators and lab technicians can standardise their in-vitro workflows.

Every protocol in this document is framed for **in-vitro research use only**. No section describes, recommends, or implies use of any compound in humans or animals. Where reconstitution volumes are given, they refer to preparing stock solutions at defined molar or mass/volume concentrations for bench work.

How to use: chapters 2–8 are foundational and apply to every compound. Compound-specific chapters (9–21) assume you have already read the foundational material. Chapters 22–26 are reference material for day-to-day lab operations.

Laboratory Note: This document does not constitute medical, veterinary, or clinical advice. Consult your institution's biosafety and ethics committees before beginning any research programme.

Chapter 2

UK Legal & Compliance Framework

Research-use classification under UK law

Research compounds supplied by PH Labs are sold strictly for laboratory research and analytical reference purposes. They are not medicinal products within the meaning of the Human Medicines Regulations 2012 and are not authorised for administration to humans or animals.

By purchasing from PH Labs, the buyer confirms they are a bona fide researcher, laboratory, academic institution, or industrial R&D facility, and that the compounds will be used exclusively for in-vitro or analytical work in accordance with UK law.

Buyers are responsible for ensuring compliance with the Control of Substances Hazardous to Health (COSHH) Regulations 2002, their institution's biosafety framework, and any local waste-disposal requirements.

Classification	Detail
Intended use	In-vitro research and analytical reference only
Not classified as	Medicinal product, food, supplement, cosmetic
Buyer requirement	18+, bona fide research context
Record keeping	COSHH assessment recommended per compound

Chapter 3

Laboratory Safety & PPE

Personal protective equipment and handling

Lyophilised peptides and small-molecule research compounds should be handled with the same care as any bioactive laboratory reagent. Always consult the compound's Safety Data Sheet (SDS) before opening the vial.

Minimum PPE when weighing, reconstituting, or aliquoting: nitrile gloves, safety glasses, and a lab coat. Work in a certified fume hood or biosafety cabinet where local risk assessment requires it.

Avoid generating aerosols. Peptide powders can be electrostatic; open vials slowly and allow them to reach room temperature before breaking the seal to minimise condensation and moisture uptake.

Hazard	Control
Skin contact with powder	Nitrile gloves, immediate wash if exposed
Inhalation of dust	Weigh in a fume hood; keep vial capped when idle
Cross-contamination	Single-use spatulas or sterile transfer needles
Sharps (reconstitution)	Sharps bin; never re-cap needles

Laboratory Note: All work with unknown-hazard compounds should default to the highest reasonable PPE tier until a COSHH assessment is complete.

Chapter 4

Reconstitution Fundamentals

Preparing stock solutions for bench use

Reconstitution converts lyophilised peptide into a defined-concentration stock solution for analytical or in-vitro assays. The two variables that matter most are **solvent choice** and **injection technique**.

For most peptides in this catalogue, bacteriostatic water (0.9% benzyl alcohol) is the default solvent for research stock solutions intended for in-vitro use over days to weeks. Sterile water for injection can be used for single-day preparations. Some hydrophobic peptides require a small percentage of acetic acid or DMSO as a co-solvent — refer to the compound-specific chapter.

Always add solvent slowly down the inside wall of the vial. Do not inject directly onto the powder pellet, as this fractures it and generates foam, which can denature the peptide. Once solvent is added, swirl gently — **never shake**.

Step	Action
1	Bring vial and solvent to room temperature (20–25 °C)
2	Wipe both septa with 70% IPA and allow to dry
3	Draw the calculated solvent volume with a sterile syringe
4	Add solvent slowly down the inside wall of the vial
5	Swirl gently until fully dissolved (do not shake)
6	Label vial: compound, concentration, solvent, date, initials

Chapter 5

Bacteriostatic Water — Solvent Guide

The default solvent for multi-day research stocks

Bacteriostatic water is sterile water containing 0.9% benzyl alcohol as a bacteriostatic preservative. The preservative inhibits microbial growth in the reconstituted stock, extending its usable bench life for research work.

Bacteriostatic water is preferred over sterile water when a stock will be sampled repeatedly over days to weeks. For assays where benzyl alcohol may interfere with the readout (e.g. certain cell culture systems), sterile water for injection or an assay-specific buffer should be used instead.

Property	Value
Composition	Sterile H2O + 0.9% benzyl alcohol
Typical pH	4.5 – 7.0
Storage (unopened)	Room temperature, out of direct light
Storage (opened multi-dose)	Refrigerate 2–8 °C, discard after 28 days
Compatible pairings	Most peptides in this catalogue

Laboratory Note: Discard any vial with visible particulates, cloudiness, or a broken seal.

Chapter 6

Storage & Cold-Chain Best Practices

Maximising stability of lyophilised and reconstituted peptides

Peptide stability is governed by three factors: **temperature**, **moisture**, and **light exposure**. Correctly stored lyophilised peptides remain stable for 24+ months; the same peptide reconstituted and left at room temperature can degrade in hours.

The table below is a general reference. Compound-specific chapters override these defaults where a peptide has unusual stability characteristics.

State	Temperature	Typical Stability
Lyophilised, sealed	–20 °C, desiccated, dark	24+ months
Lyophilised, sealed	2–8 °C, desiccated, dark	3–6 months
Reconstituted (bac. water)	2–8 °C, dark	2–6 weeks (compound-dependent)
Reconstituted, aliquoted	–20 °C, single freeze/thaw	3–6 months
Reconstituted	Room temperature	Hours — use same day

Laboratory Note: Avoid repeated freeze/thaw cycles. Aliquot the reconstituted stock into single-use volumes before freezing.

Chapter 7

Certificate of Analysis (CoA) Interpretation

Reading the QC document that ships with every batch

Every batch shipped by PH Labs is accompanied by a Certificate of Analysis (CoA) documenting identity, purity, and physical characteristics. The CoA is your primary QC document and should be filed alongside the vial in your laboratory records.

A typical CoA includes: batch/lot number, compound name and molecular formula, molecular weight, HPLC purity (%), mass spec confirmation, appearance, and manufacture date. Reference the CoA batch number in every experimental notebook entry so that results are reproducible against a specific lot.

Field	What it tells you
Batch / Lot #	Unique identifier for traceability
HPLC purity	% of target compound vs. impurities (aim ≥ 98%)
Mass spec (m/z)	Confirms molecular identity
Appearance	Expected physical state (e.g. white lyophilised powder)
Manufacture date	Starting point for the 24-month stability window

Chapter 8

HPLC & Mass Spectrometry — Reading the Data

Interpreting the analytical traces on your CoA

High-Performance Liquid Chromatography (HPLC) separates compounds in a sample by their affinity for a stationary phase. On a CoA, the HPLC trace shows a dominant peak (the target peptide) and any smaller peaks representing impurities or synthesis by-products. The area-under-curve of the main peak, expressed as a percentage of the total peak area, gives the reported purity value.

Mass spectrometry confirms that the dominant HPLC peak is in fact the intended peptide by measuring its mass-to-charge ratio (m/z). The observed m/z should match the theoretical monoisotopic mass of the target sequence to within instrument tolerance.

A CoA showing ≥ 98% HPLC purity and a matching m/z is the industry benchmark for research-grade peptides.

Laboratory Note: If a CoA reports purity below 95%, consider whether the impurity profile is compatible with your assay before proceeding.

Chapter 9

BPC-157 — Research Protocol

Body Protection Compound 157 — a synthetic pentadecapeptide

BPC-157 is a stable synthetic pentadecapeptide derived from a partial sequence of body protection compound found in gastric juice. It is widely used as a reference compound in in-vitro angiogenesis, fibroblast migration, and tendon explant research models.

Property	Value
Sequence	GEPPPGKPADDAGLV (15 aa)
Molecular formula	$C_{62}H_{98}N_{16}O_{22}$
Molecular weight	~1419.5 g/mol
Appearance	White lyophilised powder
Typical HPLC purity	≥ 99%

Laboratory Note: For in-vitro research use only. Not for human or veterinary administration.

Reconstitution reference

Parameter	Detail
Recommended solvent	Bacteriostatic water
Typical stock concentration	1–5 mg/mL
Foaming risk	Low — swirl gently
Post-reconstitution stability (2–8 °C)	~4 weeks

Storage: Lyophilised: –20 °C, desiccated, dark, 24+ months. Reconstituted: 2–8 °C, dark, use within 4 weeks; aliquot for longer storage.

Laboratory notes: Frequently used alongside TB-500 in comparative in-vitro migration assays.

Chapter 10

TB-500 — Application Guide

A synthetic 17-residue fragment of Thymosin β-4

TB-500 is a synthetic peptide corresponding to the active region of Thymosin β-4 (Tβ4). It is used in vitro to investigate G-actin sequestration, cell migration, and angiogenesis pathways.

Property	Value
Sequence	LKKTETQ (active region, 17-mer synthetic)
Molecular weight	~888 g/mol (fragment)
Appearance	White lyophilised powder
Typical HPLC purity	≥ 98%

Laboratory Note: For in-vitro research use only. Not for human or veterinary administration.

Reconstitution reference

Parameter	Detail
Recommended solvent	Bacteriostatic water
Typical stock concentration	1–2 mg/mL
Co-solvent needed	No
Post-reconstitution stability (2–8 °C)	~4 weeks

Storage: Lyophilised: –20 °C, desiccated. Reconstituted: 2–8 °C, dark.

Chapter 11

GHK-Cu — Handling & Reconstitution

Copper-tripeptide-1, a naturally occurring peptide-copper complex

GHK-Cu is a tripeptide (glycyl-L-histidyl-L-lysine) chelated to copper(II). It is used in vitro to study extracellular matrix remodelling, fibroblast activity and copper-dependent signalling.

Property	Value
Sequence	Gly-His-Lys · Cu2■
Molecular weight	~340 g/mol (Cu complex)
Appearance	Blue lyophilised powder
Typical HPLC purity	≥ 98%

Laboratory Note: For in-vitro research use only. Not for human or veterinary administration.

Reconstitution reference

Parameter	Detail
Recommended solvent	Sterile water or bacteriostatic water
Typical stock concentration	1–10 mg/mL
Light sensitivity	Moderate — store in amber vial
Post-reconstitution stability (2–8 °C)	2–4 weeks

Storage: Lyophilised: –20 °C, desiccated, protected from light.

Laboratory notes: Blue colour is expected — it reflects the Cu2■ chelation state.

Chapter 12

KPV — Research Notes

C-terminal tripeptide of α-MSH (Lys-Pro-Val)

KPV is the C-terminal tripeptide of α-melanocyte-stimulating hormone. It is widely used in vitro in inflammation and intestinal-barrier research models.

Property	Value
Sequence	Lys-Pro-Val
Molecular weight	~342 g/mol
Appearance	White lyophilised powder
Typical HPLC purity	≥ 98%

Laboratory Note: For in-vitro research use only. Not for human or veterinary administration.

Reconstitution reference

Parameter	Detail
Recommended solvent	Bacteriostatic water
Typical stock concentration	1–5 mg/mL
Post-reconstitution stability (2–8 °C)	~4 weeks

Storage: Lyophilised: –20 °C, desiccated, dark.

Chapter 13

MOTS-c — Handling

A 16-residue mitochondrial-derived peptide

MOTS-c (Mitochondrial Open Reading frame of the Twelve S rRNA type-c) is a 16-amino-acid peptide encoded within the mitochondrial genome. It is used in vitro to study metabolic and mitochondrial signalling pathways.

Property	Value
Sequence	MRWQEMGYIFYPRKLR (16 aa)
Molecular weight	~2174 g/mol
Appearance	White lyophilised powder
Typical HPLC purity	≥ 98%

Laboratory Note: For in-vitro research use only. Not for human or veterinary administration.

Reconstitution reference

Parameter	Detail
Recommended solvent	Bacteriostatic water
Typical stock concentration	1–2 mg/mL
Post-reconstitution stability (2–8 °C)	~3 weeks

Storage: Lyophilised: –20 °C, desiccated, dark.

Chapter 14

PT-141 — Handling

A cyclic heptapeptide analog of α-MSH (Bremelanotide)

PT-141 is a synthetic cyclic peptide melanocortin receptor agonist. In research contexts it is used to study melanocortin receptor pharmacology in vitro.

Property	Value
Molecular formula	$C_{50}H_{68}N_{14}O_{10}$
Molecular weight	~1025 g/mol
Appearance	White to off-white lyophilised powder
Typical HPLC purity	≥ 98%

Laboratory Note: For in-vitro research use only. Not for human or veterinary administration.

Reconstitution reference

Parameter	Detail
Recommended solvent	Bacteriostatic water
Typical stock concentration	1–5 mg/mL
Post-reconstitution stability (2–8 °C)	~4 weeks

Storage: Lyophilised: –20 °C, desiccated, dark.

Chapter 15

Melanotan-II — Handling

A cyclic heptapeptide α-MSH analog

Melanotan-II (MT-II) is a synthetic analog of α-melanocyte-stimulating hormone used in vitro to study melanocortin receptor activity and melanogenic pathways in cultured melanocyte lines.

Property	Value
Molecular formula	$C_{50}H_{69}N_{15}O_9$
Molecular weight	~1024 g/mol
Appearance	White lyophilised powder
Typical HPLC purity	≥ 98%

Laboratory Note: For in-vitro research use only. Not for human or veterinary administration.

Reconstitution reference

Parameter	Detail
Recommended solvent	Bacteriostatic water
Typical stock concentration	1–10 mg/mL
Post-reconstitution stability (2–8 °C)	~4 weeks

Storage: Lyophilised: –20 °C, desiccated, dark.

Chapter 16

Semaglutide — Research Handbook

A long-acting GLP-1 receptor agonist analog

Semaglutide is a 31-amino-acid GLP-1 receptor agonist analog. In research it is used in vitro to study GLP-1 receptor signalling, insulin-secretion assays and receptor-binding kinetics in cell-based systems.

Property	Value
Molecular weight	~4113 g/mol
Appearance	White to off-white lyophilised powder
Typical HPLC purity	≥ 98%
Solubility	Bacteriostatic water; slightly hygroscopic

Laboratory Note: For in-vitro research use only. Not for human or veterinary administration.

Reconstitution reference

Parameter	Detail
Recommended solvent	Bacteriostatic water
Typical stock concentration	1–5 mg/mL
Foaming risk	Moderate — swirl very gently
Post-reconstitution stability (2–8 °C)	~4 weeks

Storage: Lyophilised: –20 °C, desiccated, dark, 24+ months.

Chapter 17

Tirzepatide — Research Notes

A dual GIP / GLP-1 receptor agonist analog

Tirzepatide is a 39-amino-acid synthetic peptide that activates both the GIP and GLP-1 receptors. In vitro it is used in comparative receptor-agonism assays and dual-pathway signalling studies.

Property	Value
Molecular weight	~4813 g/mol
Appearance	White lyophilised powder
Typical HPLC purity	≥ 98%

Laboratory Note: For in-vitro research use only. Not for human or veterinary administration.

Reconstitution reference

Parameter	Detail
Recommended solvent	Bacteriostatic water
Typical stock concentration	1–5 mg/mL
Post-reconstitution stability (2–8 °C)	~4 weeks

Storage: Lyophilised: –20 °C, desiccated, dark.

Chapter 18

Retatrutide — Research Notes

An investigational triple GIP/GLP-1/glucagon receptor agonist

Retatrutide is a synthetic peptide investigated as a triple agonist at the GIP, GLP-1, and glucagon receptors. It is used in vitro to study multi-receptor pharmacology and comparative signalling profiles against single- and dual-agonist references.

Property	Value
Appearance	White lyophilised powder
Typical HPLC purity	≥ 98%
Solvent compatibility	Bacteriostatic water

Laboratory Note: For in-vitro research use only. Not for human or veterinary administration.

Reconstitution reference

Parameter	Detail
Recommended solvent	Bacteriostatic water
Typical stock concentration	1–5 mg/mL
Post-reconstitution stability (2–8 °C)	~4 weeks

Storage: Lyophilised: –20 °C, desiccated, dark.

Chapter 19

NAD+ — Research Compound Guide

Nicotinamide adenine dinucleotide (oxidised form)

NAD+ is an essential coenzyme used across biochemistry and cell-biology research to study redox reactions, sirtuin activity, PARP signalling, and cellular metabolism in vitro.

Property	Value
Molecular formula	$C_{21}H_{27}N_7O_{14}P_2$
Molecular weight	~663.4 g/mol
Appearance	White to off-white powder
Typical HPLC purity	≥ 98%

Laboratory Note: For in-vitro research use only. Not for human or veterinary administration.

Reconstitution reference

Parameter	Detail
Recommended solvent	Sterile water or PBS
Typical stock concentration	10–100 mM
Post-reconstitution stability (2–8 °C)	~1 week; aliquot & freeze for longer

Storage: Lyophilised: –20 °C, desiccated, dark. Reconstituted: aliquot and store at –20 °C to –80 °C; avoid freeze/thaw cycles.

Chapter 20

GLOW Blend — Handling

Research blend containing GHK-Cu, BPC-157 and TB-500

GLOW is a research blend combining GHK-Cu, BPC-157 and TB-500 in a single vial for comparative in-vitro work on wound-healing and matrix-remodelling pathways where these three references are commonly studied together.

Property	Value
Composition	GHK-Cu + BPC-157 + TB-500
Appearance	Pale-blue lyophilised powder (from GHK-Cu)

Laboratory Note: For in-vitro research use only. Not for human or veterinary administration.

Reconstitution reference

Parameter	Detail
Recommended solvent	Bacteriostatic water
Typical stock concentration	As per per-component mass on CoA
Post-reconstitution stability (2–8 °C)	2–3 weeks

Storage: Lyophilised: –20 °C, desiccated, dark.

Laboratory notes: Refer to the batch-specific CoA for the exact mass of each component.

Chapter 21

KLOW Blend — Handling

Research blend containing KPV, GHK-Cu, BPC-157 and TB-500

KLOW extends the GLOW blend by adding KPV. It is used as a comparative reference in in-vitro inflammation and barrier-integrity research models.

Property	Value
Composition	KPV + GHK-Cu + BPC-157 + TB-500
Appearance	Pale-blue lyophilised powder

Laboratory Note: For in-vitro research use only. Not for human or veterinary administration.

Reconstitution reference

Parameter	Detail
Recommended solvent	Bacteriostatic water
Typical stock concentration	As per per-component mass on CoA
Post-reconstitution stability (2–8 °C)	2–3 weeks

Storage: Lyophilised: –20 °C, desiccated, dark.

Chapter 22

Solubility & Concentration Reference

Quick-lookup solvent compatibility for the current PH Labs catalogue.

Compound	Preferred Solvent	Notes
BPC-157	Bacteriostatic water	Highly soluble, low foaming
TB-500	Bacteriostatic water	Soluble
GHK-Cu	Sterile / bac. water	Solution is pale blue
KPV	Bacteriostatic water	Highly soluble
MOTS-c	Bacteriostatic water	Soluble; keep cold
PT-141	Bacteriostatic water	Soluble
Melanotan-II	Bacteriostatic water	Soluble
Semaglutide	Bacteriostatic water	Moderate foaming — swirl slowly
Tirzepatide	Bacteriostatic water	Soluble
Retatrutide	Bacteriostatic water	Soluble
NAD+	Sterile water / PBS	Avoid alkaline solvents

Concentration formula: C (mg/mL) = mass in vial (mg) ÷ solvent volume added (mL). Record both values in your lab notebook alongside the batch number from the CoA.

Chapter 23

Lab Documentation & Record Keeping

What to record and where

Reproducibility depends on records. Every vial reconstituted in your lab should generate a corresponding notebook entry, and every experiment should reference that entry by ID so results can be traced back to a specific batch and preparation.

A minimum notebook entry for a reconstituted stock should include: date, operator initials, compound name, PH Labs batch/lot number, solvent used, solvent volume, resulting concentration, storage location, and a unique internal stock ID.

Field	Example
Date	2026-07-08
Compound / Batch	BPC-157 / Lot 26-0412
Solvent	Bacteriostatic water
Volume added	2.0 mL
Resulting concentration	5 mg/mL
Storage	Fridge 2, shelf 3, box A
Stock ID	PHL-BPC-260708-01

Chapter 24

Troubleshooting Common Issues

Symptoms, likely causes and corrective actions

The most common issues encountered during peptide reconstitution and storage are usually straightforward once you can match a symptom to its likely cause. The table below covers the top-reported issues.

Symptom	Likely cause / action
Cloudy solution after reconstitution	Solvent added too quickly; particulates; discard if persistent
Foam that will not settle	Shaking instead of swirling; wait 20 min, do not centrifuge harshly
Powder will not dissolve	Wrong solvent; check compound-specific chapter for co-solvent needs
Powder stuck to septum	Vial was inverted before reconstitution; tap gently, then reconstitute
Colour change on storage	Oxidation or light exposure; store in amber vial at –20 °C
Assay results drifting week-to-week	Freeze/thaw damage; aliquot next batch into single-use volumes

Chapter 25

Frequently Asked Questions

Are PH Labs compounds intended for human use?

No. All PH Labs research compounds are supplied strictly for in-vitro research and analytical reference. They are not medicinal products.

Do you provide dosing instructions?

No. Because our compounds are not for human or veterinary use, we do not publish dosing guidance. Reconstitution guidance in this document is for preparing bench stock solutions at defined concentrations.

What HPLC purity should I expect?

All compounds ship with a Certificate of Analysis. Typical purity is ≥ 98%, with most peptides at ≥ 99%. Check the specific CoA for each lot.

Can I store reconstituted peptide long-term?

Yes — aliquot into single-use volumes and store at –20 °C. Avoid repeated freeze/thaw cycles, which are the single largest driver of peptide degradation in a lab setting.

Do you supply bacteriostatic water?

Yes — bacteriostatic water is available as a research compound in the PH Labs catalogue.

What if I receive a damaged vial?

Do not use it. Photograph the vial and email support@phlabs.co.uk within 7 days of delivery with your order number.

Chapter 26

Contact, Support & Further Reading

PH Labs is a UK-based supplier of HPLC-verified research compounds. Our lab and customer-support team are available for technical questions about handling, reconstitution, storage and Certificate of Analysis interpretation.

Support & technical questions

Email: info@phlabs.co.uk

Website: phlabs.co.uk

Research Hub: phlabs.co.uk/resources

Further reading

- • PH Labs Research Compound Catalogue — full product listing
- • Peptide reference — reconstitution volumes and syringe unit conversions in phlabs.co.uk/resources
- • Certificate of Analysis library — batch-specific CoAs available on request

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